

Junaid Rehman

Neuromorphic Research Engineer | Islamabad, Pakistan

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Education

B.S. Electrical Engineering (Electronics)

2017 – 2021

Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad

Thesis: Fabrication and Characterization of FET and Diode Devices Using Transition Metal Dichalcogenides (TMDs)

- Relevant Coursework: Real time embedded system, Integrated Electronics, Digital Signal Processing, Semiconductor Devices

Research Experience

Neuromorphic Research Engineer

2025 – Present

Project Phasor (Remote)

- Architecting hardware-aware spiking neural networks (SNNs) that translate cortical column dynamics into silicon-ready models; co-designing algorithms with analog/mixed-signal hardware constraints.
- Engineering engram-based memory layers using PyTorch/snntorch to achieve sub-microwatt inference targets for edge neuromorphic deployment.
- Conducted five-model ablation study (LIF, SCL, Hopfield, HGRN, Full Hybrid) on N-MNIST; achieved synergistic balance (97.49% accuracy, 1.85 μ J inference, 97% spike sparsity).
- **Current:** Porting cross-modal HGRN architecture to Intel Lava and sPyNNaker to evaluate deployment constraints on Intel Loihi and SpiNNaker2 multicast fabrics.

Analog / Mixed-Signal Design Research Assistant

2022 – 2023

GSME, Islamabad

- Designed a 65 dB gain operational amplifier in skywater 130nm; full GDSII layout with advanced matching techniques, DRC/LVS clean.
- Designed 200 MHz PLL with wide tuning range for high-speed ADC/DAC clocking; post-layout simulation including parasitic extraction (PEX).
- Developed delay-free insensitive switched-capacitor integrator and non-overlapping clock generator (40 ns period, 5 ns dead-time).
- Tools: Cadence Virtuoso, Spectre, Xschem, Netgen, Calibre; TSMC 65nm, SKY130, GF180 PDKs.

Publications

Modality-Dependent Memory Mechanisms in Cross-Modal Neuromorphic Computing

E. Blessing, **J. Rehman**, S. Roy, M. Kim, C. Tseng, I. Nkrumah

IEEE Computer, Special Issue on Neuromorphic Computing, 2026

- First cross-modal ablation study of memory mechanisms in SNNs across visual (N-MNIST, DVS-Gesture, Caltech101-DVS) and auditory (SHD, SSC) neuromorphic datasets.
- Revealed severe modality-dependent specialization: Hopfield networks achieve 97.68% on visual tasks but collapse to 76.15% on auditory tasks (21.53-point gap), exposing fundamental incompatibility between energy-based associative memory and temporal sequential processing.
- Full Hybrid (M5) architecture resolves the bottleneck via HGRN temporal gating, achieving peak cross-modal average of 89.99% (97.72% visual, 82.25% audio) — best across all five architectures.
- Optimized HGRN (M4) achieves 719 $\times$ energy efficiency over dense ANNs at 99.3% spike sparsity; Hopfield memory shown compatible with memristor crossbar implementations, HGRN gating with analog mixed-signal circuits.

Memory-Augmented Spiking Networks: Synergistic Integration of Complementary Mechanisms for Neuromorphic Vision

E. Blessing, C. Tseng, I. Nkrumah, **J. Rehman**

Neuro-Inspired Computational Elements Conference (NICE 2026) — Proceedings Paper & Poster Presentation

- Five-model ablation study integrating LIF neurons, SCL, Hopfield networks, and HGRN on N-MNIST, revealing that architectural balance outperforms individual component optimization.
- Discovered unexpected trade-offs: SCL improves accuracy (+0.28%) but disrupts spike-timing clustering (0.687 → 0.637 silhouette); HGRN provides consistent gains across all metrics (+1.01% accuracy, 170.6× energy efficiency).
- Full hybrid achieves 97.49% accuracy at 1.85 μJ with 97.0% spike sparsity and silhouette 0.715 — exceeding the “excellent” clustering threshold through synergistic integration.

Industry Experience

Co-founder

2025 – Present

Two Apps, Dubai UAE (Remote)

- Co-founded AI automation agency delivering agentic AI workflows and white-label AI implementation for businesses globally.

AI Engineer

2022 – 2024

IdrakAI, Islamabad

- Deployed production OCR pipelines processing 10,000+ daily documents for banking verification; reduced latency to 0.1 s via chunking and INT8 quantization.
- Fine-tuned TinyLlama and Llama 3 using ORPO for domain-specific tasks; deployed on-device via Ollama + LangChain.

Technical Skills

Neuromorphic:	SNNs, LIF neurons, Event-based Vision, snntorch, Norse, Lava (Intel Loihi), sPyNNaker, Surrogate Gradients, STDP
Analog IC:	Cadence Virtuoso, Spectre, GDSII Layout, PLLs, Op-Amps, SC Circuits, PEX
Digital:	Verilog, SystemVerilog, RTL Design, Synthesis, STA, FPGA Prototyping
Process Tech:	TSMC 65nm, SkyWater 130nm (SKY130)
Languages/Tools:	Python, C++, PyTorch, Docker, Git, Linux, Flutter, AWS

Selected Achievements

- National Level Rugby Player – Division 1 league, Pakistan
- Languages: English (C1 Professional Proficiency) | Urdu (Native) | Arabic (Basic)